Adam Awale
Tucson, AZ | (520) 449-9659 | awaleadam@gmail.com | https://github.com/adamawale/projects | https://www.linkedin.com/in/adam-awale

EDUCATION

Arizona State University Phoenix, AZ
PhD Student in Computer Engineering, GPA: 3.78 *Aug 2021 - Present*

University of Arizona Tucson, AZ
Master of Science in Electrical and Computer Engineering *Dec. 2019*
Relevant Courses: Software Engineering, Machine Learning, Networking, Control Systems, Signal Processing

University of Arizona Tucson, AZ
Bachelor of Science in Electrical and Computer Engineering
Minor in Computer Science, Minor in Mathematics *May 2018*
Relevant Courses: Algorithm, Data Structures, Computer Architecture

SKILLS

- Python, Matlab/simulink, ROS, LabView, Tensorflow, C++, C, Java, Javascript, Unix, Bash, Ruby, html, CSS, SQL, PHP, Assembly, Verilog, Pspice

WORK EXPERIENCE

Arizona State University Phoenix AZ
Research assistant *Aug. 2021 – July 2022*
- Participated in research on secure scalable RISC-V HPC systems
- Developed evaluation tools and simulators for ongoing RISC-V projects
- Developed automated validation tests for our RISC-V project which used fuzzing to find edge cases

Arizona State University Phoenix AZ
Teaching assistant *Jan. 2022 – May 2022*
- Assisted in teaching “Computer Architecture CSE 420” course where we focused on RISC-V architecture
- Created the exams, homework, and labs for the class, and graded them
- Led recitation hours where I would provide supplemental lectures to students that fell behind

General Motors Tucson AZ
Software Developer *Nov. 2020 – Aug. 2021*
- Automated integration of new and existing Programmable Logic Controllers (PLCs)
- Implemented addition of maintenance calls to existing PLCs

Photometrics Tucson AZ
Research Intern *June 2019 – Sept. 2019*
- Autofocused scientific cameras using artificial intelligence and real time inference on ARM architecture
- Optimized the convolutional neural network using data structures and reduced I/O overhead on the inference loop
- Wrote C++ .so files to interface imaging API with Python modules
- Wrote I2C protocols for devices in C++ to interface with Linux and Windows systems for real time inference loop

Microchip San Jose, CA
Research Intern *June 2018 – Sept. 2018*
- Automated extraction of a matrix of statistics for all FPGA designs, improving time to analyze new and current designs by 80%
- Implemented a machine learning model that flagged potential net delay errors for more in depth verification instead of analyzing everything
- Defined requirements and parameters for this model and problem statement
- Verified validity of various models in SkLearn, Keras, Tensorflow to determine best fit with requirements.

Microsemi San Jose, CA
Research Intern *June 2017 – Sept. 2017*
- Automated FPGA core update processes to decrease update times by 15%
- Automated extraction of timing and power metrics to increase extraction speed by 70%
- Ran Regression tests on Linux servers using Bash scripts for the newest generation of FPGAs

Research Intern *June 2016 – Sept. 2016*
- Extracted data from dump files for the newest generation of FPGAs.
- Analyzed this data to identify trends and future errors to increase error detection speed by 30%

University of Arizona Tucson, AZ
Teaching Assistant *Dec. 2016 – May 2017*
- Assisted in teaching “Introduction to C” course, led lab sessions on C exercises, graded assignments and tutored students